

Voice-Based AI Receptionist Agent for Smart College Communication

Using RAG and Agentic AI

Project Proposal — Mentor Discussion Document

Rajalakshmi Engineering College | B.E. Computer Science and Engineering

Abstract

Educational institutions receive numerous telephone inquiries every day regarding admissions, courses, departments, fee structure, placements, hostel facilities, academic schedules, and student services. Handling every call manually increases the workload of reception staff, leads to longer waiting times, and may result in inconsistent responses. To address these challenges, this project proposes a Voice-Based AI Receptionist Agent capable of answering calls autonomously using speech recognition, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), and Agentic AI.

In Phase 1, the voice agent is trained on publicly available college information such as admissions, departments, faculty details, campus facilities, placements, events, and academic regulations. Incoming calls are first handled by the AI receptionist, which converts speech to text, retrieves relevant information from the institutional knowledge base, generates an accurate response, and communicates it back to the caller using text-to-speech. The agent is designed to provide only information grounded in the college knowledge base and politely decline or seek clarification when sufficient information is unavailable.

In Phase 2, the system is extended with an authenticated Agentic AI layer that securely integrates with the college's ERP or DigiCampus platform to answer personalized student queries such as attendance, internal marks, fee status, examination results, and placement eligibility. If a caller asks a question that requires human judgment, administrative approval, or cannot be resolved confidently by the AI, the call is automatically transferred to the human receptionist, ensuring seamless escalation and improved service quality.

The proposed solution aims to reduce receptionist workload, provide instant and accurate responses, improve caller experience, ensure 24×7 availability, and enable efficient management of college inquiries while maintaining security, privacy, and reliable information delivery.

1. Background and Motivation

College reception desks are a critical point of contact for prospective students, parents, and existing students, yet they operate under significant resource constraints. A single receptionist handles a mix of walk-in visitors, email inquiries, and telephone calls simultaneously, often during high-pressure periods such as admissions season, exam schedules, and placement drives. The result is long hold times, inconsistent information delivery depending on which staff member takes a call, and complete unavailability outside office hours.

The Rajalakshmi Engineering College (REC) website already includes an entry point for an AI assistant (“Ask Aadhi” on the homepage), which demonstrates institutional readiness for AI-assisted interaction. This project takes that concept further by targeting the telephone communication channel — the highest-volume, highest-friction interaction mode — and replacing or augmenting it with an intelligent voice agent that can handle the majority of common queries autonomously, escalating to a human only when genuinely needed.

2. Problem Statement

The college currently has no automated voice system capable of understanding natural-language telephone queries and providing accurate, context-grounded responses drawn from institutional data. Specific problems this project addresses:

- High call volume to reception staff for routine, repeatable queries (fee structure, admission cut-offs, hostel availability, department contacts) that do not require human judgment.
- No out-of-hours coverage: calls during evenings, weekends, and holidays go unanswered.
- Inconsistency in responses: different staff members may provide different answers to the same question depending on their knowledge and access to current documents.
- No integration between telephone inquiries and the college’s existing ERP (DigiCampus) data, meaning students cannot confirm their own attendance, results, or fee status via a phone call without being transferred to multiple departments.

Phase 1 Problem: Build a voice AI that autonomously handles telephone inquiries about publicly available institutional information (admissions, departments, faculty, facilities, placements, events, regulations) using speech recognition, RAG over the college knowledge base, LLM-generated grounded responses, and text-to-speech delivery — with no access to personal student data at this stage.

Phase 2 Problem: Extend the agent to answer authenticated, personalized student queries (attendance, marks, fees, placement eligibility) by securely integrating with DigiCampus via API, and implement intelligent call escalation to a human receptionist when the query is beyond the AI’s defined scope or confidence threshold.

3. Proposed System Overview

3.1 End-to-End Call Flow (Phase 1)

The following sequence describes how an inbound call is handled by the AI receptionist in Phase 1:

1. Caller dials the college reception number. The AI agent answers automatically (via a telephony API/SIP trunk integration).
2. The agent greets the caller and listens. Speech-to-Text (STT) converts the caller’s spoken query into text in real time.
3. The text query is passed to the RAG pipeline. Relevant chunks are retrieved from the college knowledge base (website content, PDFs, circulars, handbooks).

4. The LLM synthesizes an accurate, grounded response using only the retrieved context. If no relevant context is found, the agent politely indicates it cannot assist with that query and offers to connect the caller to reception staff.
5. Text-to-Speech (TTS) converts the generated response back to natural-sounding speech and plays it to the caller.
6. The agent handles follow-up questions within the same call session, maintaining conversation history for multi-turn interactions.

3.2 Additional Behaviour in Phase 2

Phase 2 adds two capabilities on top of the Phase 1 flow:

- **Authenticated personal data queries:** After verifying caller identity (student ID + OTP or PIN via DTMF/voice), the agent can call DigiCampus APIs to retrieve and speak back the student's attendance, results, fee status, or placement eligibility.
- **Intelligent escalation:** If the query cannot be resolved by the AI (requires administrative approval, confidential data, legal/financial judgment, or falls below the confidence threshold), the agent announces the transfer and routes the call to the available human receptionist via the telephony system, passing along a transcript summary of the conversation so the receptionist is immediately informed.

4. System Architecture

4.1 High-Level Component Diagram

The system comprises five distinct layers:

Layer	Components	Role
Telephony / Voice I/O	Twilio Voice / TELNYX / SIP Trunk + WebRTC bridge	Receives inbound calls, streams audio to and from the agent backend
Speech Processing	STT: Deepgram / Google Speech-to-Text / Whisper (OpenAI) TTS: ElevenLabs / Google TTS / Amazon Polly	Converts caller speech to text and agent text back to speech
Retrieval Layer (RAG)	Embeddings model + ChromaDB/FAISS vector store populated from college website scrape and PDF documents	Retrieves the most relevant knowledge chunks for each query
Agent / Reasoning Layer	LLM (Claude/GPT) with a grounding constraint prompt + Phase 2: tool-use/function calling for DigiCampus API + confidence-based escalation logic	Synthesizes grounded responses; decides whether to answer, ask for clarification, or escalate
Backend Orchestration	Python/FastAPI WebSocket server connecting telephony audio stream	Central coordinator: manages call session state, conversation history, auth tokens (Phase 2)

	to STT → RAG → LLM → TTS pipeline	
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4.2 Escalation Logic

The agent escalates to a human receptionist when any of the following conditions are met:

- The RAG retrieval returns no sufficiently relevant context AND the query does not fall into the authenticated-data category (Phase 2).
- The caller explicitly asks to speak to a human.
- The query involves administrative decisions, complaints, or matters requiring approval.
- The Phase 2 authentication step fails after the defined number of retries.

On escalation, the system: (a) informs the caller, (b) generates a brief text summary of the conversation so far, (c) routes the call via the telephony API to the reception desk, and (d) optionally sends the summary to the receptionist's screen or email before the call connects.

5. Technology Stack

Phase 1 Stack

Component	Chosen Tool / Library	Rationale
Telephony	Twilio Programmable Voice (or Plivo as a cost-effective alternative)	Handles call reception, audio streaming, DTMF; well-documented Python/Node SDKs
Speech-to-Text	Deepgram Nova-2 or OpenAI Whisper	Deepgram offers low-latency streaming STT; Whisper is a strong offline alternative for cost control
Text-to-Speech	ElevenLabs (natural voice) or Google TTS (cost-effective)	ElevenLabs produces the most natural-sounding output; Google TTS acceptable for prototype
Content Ingestion	Python + BeautifulSoup / Playwright for web scraping; PyMuPDF for PDF parsing	Covers both dynamic JS-rendered pages and static PDF circulars from the college site
Embeddings	OpenAI text-embedding-3-small or sentence-transformers (open-source)	Open-source option removes per-call cost during development and evaluation
Vector Database	ChromaDB (Phase 1) or FAISS	Lightweight, local, sufficient for single-institution corpus; no cloud infra needed for prototype
LLM	Claude 3 Haiku/Sonnet (Anthropic API) or GPT-4o-mini	Haiku/mini: fast and cost-effective for high call volume; Sonnet for higher accuracy on complex queries
Backend	Python / FastAPI with WebSocket support	Natural pairing with the Python-based RAG/ML stack; handles real-time audio stream coordination

Session Management	In-memory conversation state per call SID (Twilio call identifier)	Maintains multi-turn history within a single call without requiring a persistent database in Phase 1
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Phase 2 Additions

Component	Chosen Tool / Library	Rationale
Agentic Orchestration	LLM native function-calling / tool-use (Claude or OpenAI) or LangGraph	Allows LLM to decide when to call DigiCampus API vs. answer from RAG context
Caller Authentication	DTMF (keypad) PIN + Student ID, or OTP via SMS (Twilio Verify)	Voice-based verification is friction-appropriate for a telephone channel; OTP adds a second factor
DigiCampus Integration	REST API calls (read-only) to DigiCampus endpoints, with per-student JWT/session token passed server-side	Never sends student credentials over voice; all data retrieval happens server-to-server
Call Transfer	Twilio <Dial> TwiML verb with pre-transfer conversation summary webhook to receptionist's endpoint	Native telephony transfer; summary delivered via webhook or SMS to the receptionist before the call connects
Logging & Audit	Structured call logs (anonymised transcript, query category, escalation flag, response time) stored in PostgreSQL or SQLite	Required for audit, tuning, and demonstrating safety/accuracy in a real institutional context

6. Knowledge Base — Data Sources and Ingestion

6.1 Phase 1 Sources (Static, Public)

- Rajalakshmi Engineering College website: all pages under About, Academics, Admissions, Departments, Faculty, Placements, Research, Facilities, Life at Rajalakshmi, and Events.
- Downloadable PDFs published on the site: admission brochures, academic handbooks, placement reports, NAAC/AQAR documents, fee notifications, exam schedules.
- Department-specific sub-pages and faculty profile pages.

6.2 Ingestion Pipeline

7. Scraper runs against the college website (BeautifulSoup for static pages; Playwright for JS-rendered pages) and extracts clean text with page-level metadata (URL, section, date if available).
8. PDF parser (PyMuPDF) extracts text from all downloadable documents.
9. Text is chunked with overlap (e.g., 512 tokens, 64-token overlap) to preserve context across chunk boundaries.

10. Chunks are embedded and stored in ChromaDB with source metadata (URL, document title, page number) for source attribution in responses.
11. A re-ingestion schedule is defined (e.g., weekly or triggered on site update) so the knowledge base stays current with new notices, updated fee structures, and academic calendars.

6.3 Phase 2 — Live Student Data (DigiCampus API)

No student data is stored in the vector database. All personalized data is fetched live at query time via secure server-to-server API calls to DigiCampus after authentication. Fields accessible via API (subject to DigiCampus granting access):

- Attendance percentage per subject and overall
- Internal assessment marks and examination results
- Fee payment status and outstanding dues
- Placement drive eligibility status

7. Pros, Cons, and Risk Assessment

7.1 Advantages

- 24x7 availability with no staffing cost for routine queries.
- Consistent, grounded answers: the agent cannot answer beyond what is in the knowledge base, eliminating the inconsistency problem of different staff giving different answers.
- Scalable: handles multiple concurrent calls without queue time.
- Measurable quality: call logs and transcripts enable ongoing accuracy monitoring and knowledge base improvement.
- Strong resume differentiation: combines telephony engineering, real-time STT/TTS pipelines, RAG, and agentic tool use — a multi-discipline system beyond a typical student project.

7.2 Challenges and Risks

- Telephony integration complexity: real inbound call handling requires a phone number, SIP trunk configuration, or institutional telecom cooperation — for a prototype, this can be simulated via Twilio's test/trial environment or a browser-based softphone.
- STT accuracy for Indian accents and domain-specific vocabulary (college/ERP terminology): needs testing and potentially custom vocabulary hints via Deepgram/Whisper config.
- Latency: the chain STT → RAG retrieval → LLM generation → TTS must complete within ~2–3 seconds to feel responsive in a phone call. This constrains model size choices and chunk retrieval count.
- Hallucination in a voice channel is more damaging than in a text chat: a caller cannot re-read a wrong answer. Grounding constraints must be strictly enforced.

- DigiCampus API access for Phase 2 depends entirely on institutional IT approval — this is outside the developer’s direct control and should be raised early with the mentor and administration.
- Privacy and regulatory considerations: call recording and transcript storage for Indian users should be handled in compliance with applicable data protection guidance; storing student data (even temporarily) requires institutional sign-off.

8. Phased Scope and Milestones

Milestone	Deliverable	Phase
M1	College website scraping + PDF ingestion pipeline complete; ChromaDB populated and searchable	1
M2	RAG backend answers test queries accurately in isolation (text-in / text-out, no voice yet)	1
M3	STT → RAG → LLM → TTS pipeline integrated end-to-end in a browser-based demo (microphone input, speaker output — no telephony yet)	1
M4	Twilio telephony integration: inbound call to a Twilio number is handled by the AI agent	1
M5	Evaluation: benchmark of 20 sample call queries (direct + scenario-based); grounding and refusal behaviour verified; latency measured	1
M6	Mentor review and decision on DigiCampus API access; institution sign-off on Phase 2 scope	Transition
M7	Caller authentication flow (Student ID + PIN/OTP) implemented and tested	2
M8	DigiCampus API integration (or sandboxed mock): attendance, marks, fee, placement eligibility queries working	2
M9	Escalation-to-human-receptionist flow implemented and tested: transfer + transcript summary delivery	2
M10	End-to-end Phase 2 evaluation; final documentation and demo recording	2

9. Open Questions for Mentor Discussion

12. Telephony integration approach: For Phase 1, is it acceptable to demo the voice pipeline via a browser-based softphone/microphone (no real phone number), or should we target a live Twilio number from the start? The latter requires a Twilio account and a modest budget for call minutes.
13. STT language and accent configuration: Should we evaluate multiple STT engines for accuracy on Tamil-English code-mixed speech (common in REC caller demographics), or start with standard English-only and adjust after testing?

14. Latency budget: What is an acceptable response delay for the voice agent from the caller's perspective — 2 seconds? 3 seconds? This decision drives whether we use a faster, smaller LLM (Haiku/mini) or a more accurate but slower one (Sonnet/GPT-4o), and how many chunks we retrieve.
15. DigiCampus API: Does the institution's IT department expose (or can it expose) a documented REST API for student data, or would Phase 2 require alternative approaches such as a read-only database connector or an intermediary microservice? This is the single largest external dependency and must be clarified before Phase 2 development begins.
16. Escalation endpoint: Does the college have a SIP-capable desk phone or PBX system that Twilio can transfer calls to, or would escalation need to use a mobile number/PSTN transfer? This affects the cost and technical complexity of the transfer feature.
17. Data retention policy: Should call transcripts and query logs be stored for quality/audit purposes, and if so, what is the permissible retention period and storage location under the college's existing IT policy?

10. Academic and Career Value of This Project

This project spans multiple technical domains rarely combined in a single undergraduate project, and is therefore a strong differentiator on a resume and in technical interviews:

Skill Domain	What Is Demonstrated
RAG Engineering	Chunking strategy, embedding selection, retrieval quality evaluation, grounding constraints
Real-Time Pipelines	STT/TTS integration, WebSocket streaming, latency-optimised API chaining
Agentic AI	LLM tool-use / function calling, routing logic, confidence-based decision making
Telephony Systems	SIP/PSTN integration, call session management, DTMF handling, call transfer
Security and Privacy	Per-user data isolation, voice-channel authentication, audit logging
Product Thinking	Escalation design, caller experience, 24x7 availability, failure modes

The project can be presented as a complete, end-to-end AI system that addresses a real institutional problem with measurable outcomes (reduction in calls handled manually, first-call resolution rate, query accuracy). This framing — real problem, real architecture, real tradeoffs — is what converts a project from a resume line item into a technical interview talking point.